Factorial Experiment Report: Effects of SO_2, Temperature, and Time on Corn Stalks

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Abstract

This project investigates the effects of three treatment factors: **sulphur dioxide (SO_2) levels**, **temperature**, and **exposure time** on the Neutral Detergent Fiber (NDF) of corn stalks. A $2 \times 2 \times 3$ factorial design with 2 replications per cell was used. ANOVA with interaction terms and Tukey HSD post-hoc tests were performed. Results show that temperature and exposure time significantly affect NDF, while SO_2 and all interactions were not significant.

1 Introduction

NDF (Neutral Detergent Fiber) is a key measure of cell wall content in plant materials. Higher NDF values generally indicate more fibrous, less digestible material.

This study investigates how NDF is affected by:

- Sulphur dioxide (SO_2) levels: 2, 4, 6 ppm
- Temperature: 20°C and 30°C
- Exposure Time: 24 and 72 hours

Each treatment combination was replicated twice (i.e., two corn stalks per combination) in a factorial $2 \times 2 \times 3$ design.

2 Model and Assumptions

Model Equation

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk} + \varepsilon_{ijkl}$$

Where:

- Y_{ijkl}: observed NDF for the l-th replicate of temperature i, time j, and SO_2 level k
- μ: overall mean

- α_i, β_j, γ_k: main effects of temperature, time, and SO_2
- $(\alpha\beta)_{ij}, (\alpha\gamma)_{ik}, (\beta\gamma)_{jk}, (\alpha\beta\gamma)_{ijk}$: interaction effects
- $\varepsilon_{ijkl} \sim N(0, \sigma^2)$: random error

Assumptions

- Errors are independent and normally distributed
- Homogeneous variances across treatment groups
- Additivity of main effects and interactions
- Random assignment of treatments

3 Data Table

Table 1: NDF values for combinations of SO_2, Temperature, and Time

Temperature (°C)	Time (hrs)	SO_2	NDF Values
20	24	2	79.4, 83.5
20	24	4	70.7, 67.6
20	24	6	59.8, 57.4
20	72	2	81.9, 81.8
20	72	4	79.2, 79.7
20	72	6	75.6, 77.1
30	24	2	69.8, 75.0
30	24	4	62.3, 66.0
30	24	6	61.9, 62.0
30	72	2	68.6, 72.0
30	72	4	66.5, 66.4
30	72	6	64.3, 62.6

4 ANOVA Results

Table 2: Three-Way ANOVA Table for NDF

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Temperature	1	386.4	386.4	10.305	0.0075 **
Time	1	339.8	339.8	9.061	0.0109 *
SO_2	2	127.7	63.8	1.702	0.223
Temperature:Time	1	33.4	33.4	0.890	0.364
Temperature:SO_2	2	6.9	3.4	0.091	0.913
Time:SO_2	2	79.6	39.8	1.061	0.376
Temperature:Time:SO_2	2	0.7	0.3	0.009	0.991
Residuals	12	449.9	37.5		

*Significance codes: ** p ¡ 0.01, * p ¡ 0.05*

5 Tukey HSD Post-hoc Test

Significant Comparisons

Temperature: 30°C has significantly lower NDF than 20°C.

Time: 72 hrs exposure has significantly lower NDF than 24 hrs.

SO_2 and all interactions: no significant differences detected.

Factor p adj	Comparison	diff	lwr	upr
Temperature 0.0075	30-20	-8.03	-13.47	-2.58
Time 0.0109	72-24	-7.53	-12.97	-2.08

Table 3: Tukey HSD pairwise comparisons for significant main effects

6 Conclusion

The factorial experiment shows that temperature and exposure time significantly influence the Neutral Detergent Fiber (NDF) content of corn stalks. Higher temperatures and longer exposure reduce NDF, suggesting a softening or breakdown of fibrous material.
SO_2 levels alone, as well as all interactions among factors, were not significant.

Overall, these results highlight that environmental conditions (temperature and duration) are the primary drivers of NDF changes in corn stalks, while SO_2 levels under the tested range do not have a measurable impact.

7 References

- Montgomery, D. C. (2017). *Design and Analysis of Experiments.* John Wiley & Sons.
- R Core Team. (2024). *R: A language and environment for statistical computing.* R Foundation for Statistical Computing, Vienna, Austria. URL: `https://www.R-project.org/`